Insurance Claims Analysis — SQL + Power BI

1. Business problem

Which policy types, customer segments and sales channels drive the highest claim ratio, and where are claims being rejected or delayed?

2. Data source

Synthetic practice dataset with 3 tables (customers, policies, claims), covering 2022-2025. It contains deliberate data-quality issues, so cleaning is a core part of the project.

Table	Raw rows	Clean rows
customers	3,060	3,000
policies	5,529	5,489
claims	1,560	1,520

3. Tools

SQL Server (SSMS) for cleaning and validation · Power BI Desktop · DAX

4. Process

4.1 Data cleaning (SQL)

Raw files were imported as text into `*_raw` tables, which were never modified. Clean tables (`dim_customers`, `dim_policies`, `fact_claims`) were built from them with SQL.

Issue found	Rows affected	Fix
Duplicate rows (customers / policies / claims)	60 / 40 / 35	Removed with `ROW_NUMBER()`
Gender stored as `F` / `M` and `Female` / `Male`	280	Standardised to Female / Male
Policy type in mixed case (`MOTOR`, `health`)	292	Standardised to title case
Claim status with mixed case and extra spaces	—	Trimmed and standardised to Approved / Rejected / Pending
Two date formats in `join_date`	1,021	Converted to one `date` type
Two date formats in `claim_date`	359	Converted to one `date` type

Issue found	Rows affected	Fix
Premium stored as text (`₹37,690`)	701	Removed symbol and commas, converted to number
Claim amount stored with commas (`37,415`)	397	Removed commas, converted to number
Invalid ages (`0`, `-1`, `150`, `999`)	25	Set to NULL (valid range 18-90)
Negative claim amounts	8	Converted with `ABS()` (after the fix, no approved amount exceeded its claim amount)
Blank claim amount / premium	40 / 50	Left as NULL, not estimated
Blank channel / occupation	134 / —	Set to `Unknown`
Claims linked to a policy that does not exist (`P999999`)	5	Excluded from analysis, saved in `claims_orphans_removed`
Claims dated outside their policy period	52	**Kept and flagged** (`outside_policy_period`)

Blank `days_to_settle` is expected for Pending and Rejected claims, so it was not treated as an error.

4.2 Data model (Power BI)

Star schema with a Date table:

- `dim_customers` (1) → (*) `dim_policies` on `customer_id`
- `dim_policies` (1) → (*) `fact_claims` on `policy_id`
- `Calendar` (1) → (*) `fact_claims` on `claim_date`

4.3 DAX measures

Total Policies, Total Premium, Total Claims, Claimed Amount, Approved Amount, **Claim Ratio** (Approved Amount ÷ Total Premium), Approved / Rejected / Pending Claims, **Approval Rate** (Approved ÷ (Approved + Rejected), pending excluded), Avg Days to Settle, Delayed Claims and Delayed % (settled in over 45 days), Claim Frequency (claims per policy), Claims LY and Claims YoY %.

4.4 Validation

Row counts were checked after cleaning (3,000 / 5,489 / 1,520), and key results were cross-checked between SQL queries and Power BI cards.

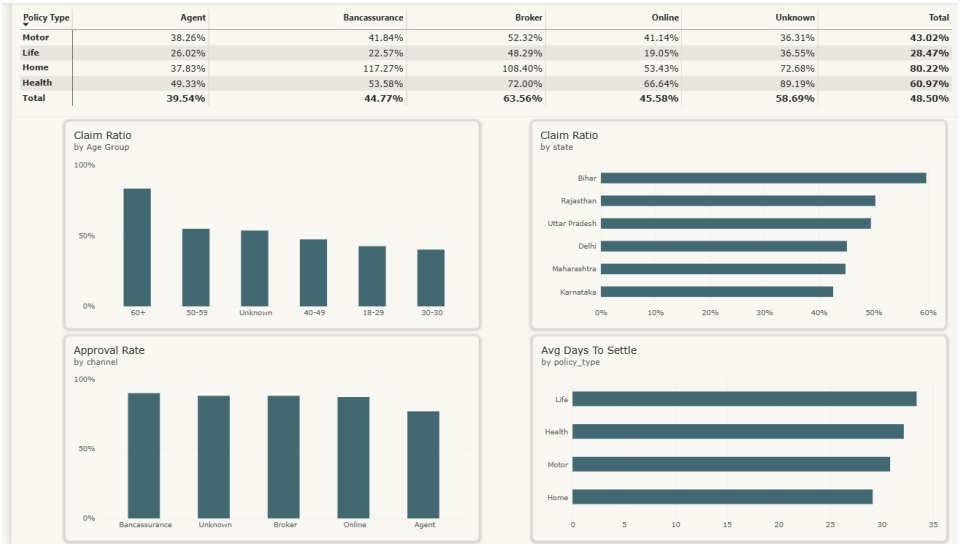
5. Key findings

Headline numbers: 5,489 policies · premium ₹111.4M · 1,520 claims · approved payouts ₹54.0M · claim ratio **48.5%** · approval rate **85.6%** · average settlement **31.6 days**.

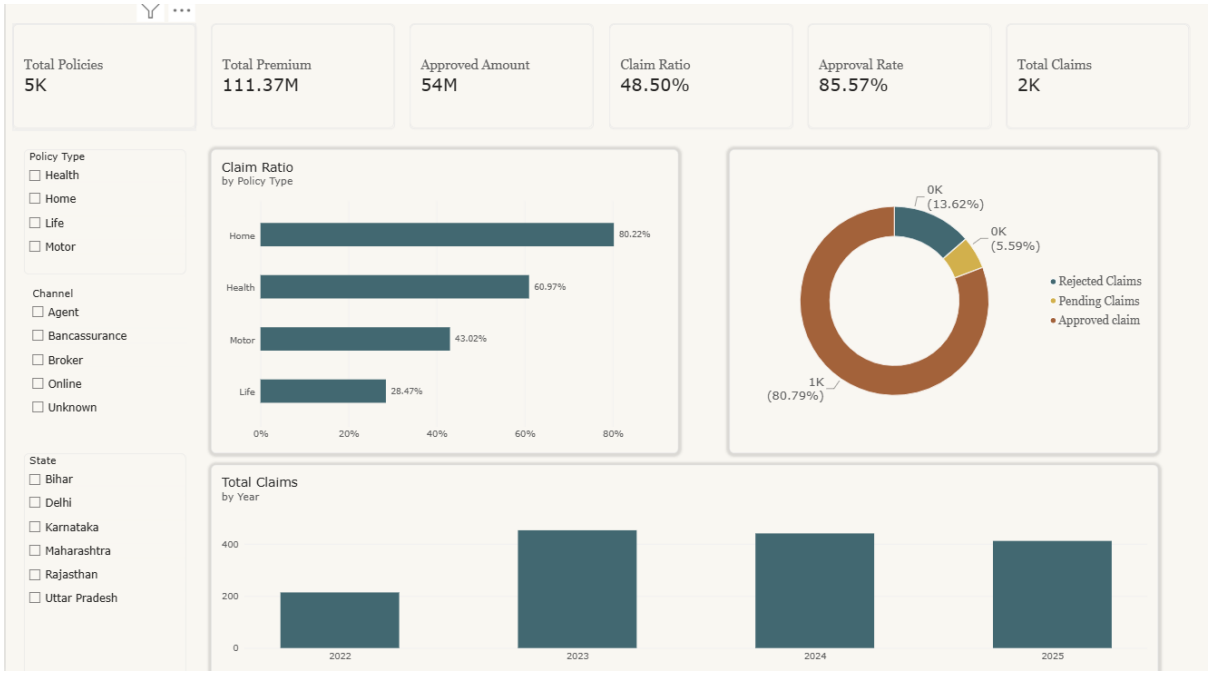
1. **Home and Health have the highest claim ratios.** Home is 80.2% and Health 61.0%, against 48.5% overall (Motor 43.0%, Life 28.5%). Health brings in about 45% of premium but about 56% of approved payouts, making it the largest source of risk. Home has only 67 claims, so that figure needs more data to confirm.
2. **Claim ratio rises sharply with age.** Customers aged 60+ claim at 83.1% of premium and the 50-59 group at 54.7%, against about 40-42% for customers under 40. Retired customers show 62.7%, versus 39.1% for self-employed customers.
3. **The Agent channel has the lowest approval rate.** Agent approval is 77.0% (84 of 366 decided claims rejected), while Bancassurance is 90.1%, Broker 88.2% and Online 87.3%.
4. **The Broker channel has the highest claim ratio.** Broker is 63.6%, against 39.5% for Agent, 45.6% for Online and 44.8% for Bancassurance. Channels differ in both approval behaviour and risk quality.
5. **About one in four approved claims is slow.** 26.1% of approved claims took more than 45 days. Life is slowest on average (33.4 days) and Home fastest (29.1 days).
6. **Claim volume has been flat to slightly falling since 2023.** Claims were 453 (2023), 441 (2024, -2.6%) and 412 (2025, -6.6%). The jump from 2022 (214 claims) reflects the policy book starting in 2022, not real growth.
7. **Gender is not a driver.** Claim ratio is 47.9% for female and 49.1% for male customers, with almost identical claim frequency.

6. Dashboard screenshots

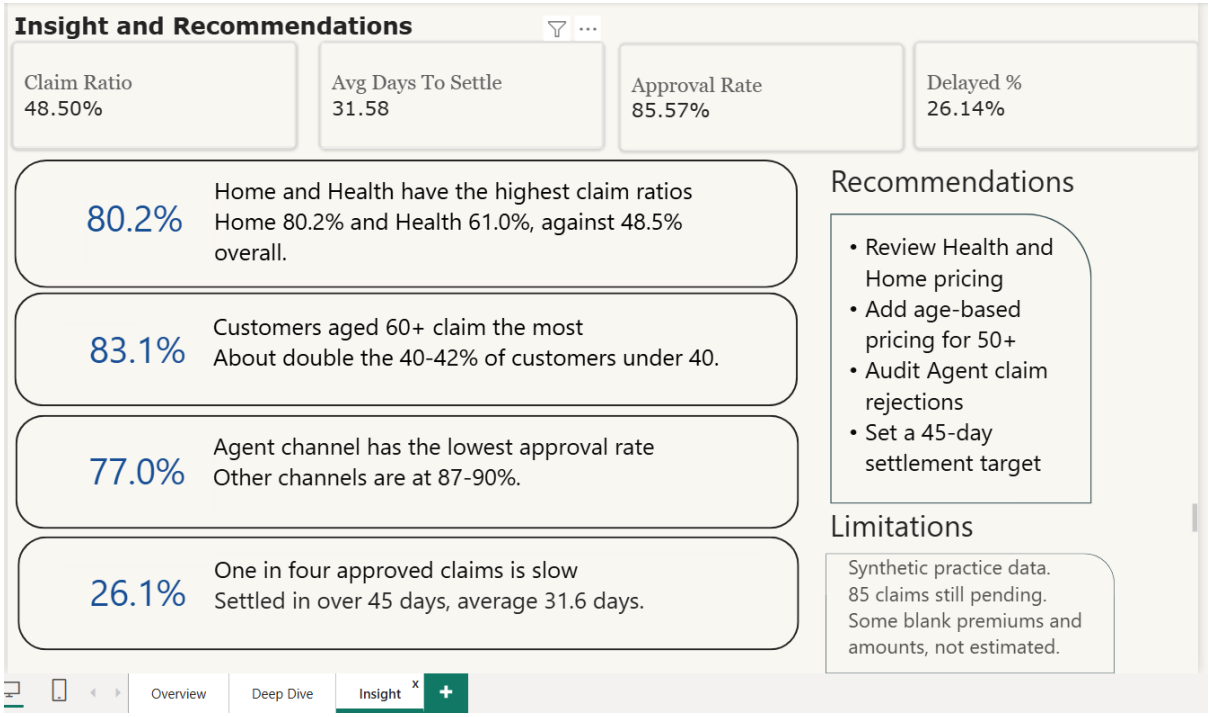
[Overview]



[Deep dive]



[Insights]



7. Recommendations

- **Health and Home:** review pricing and claim patterns; monitor Home as more data comes in.
- **Customers aged 50+:** consider age-based pricing or health-screening requirements.
- **Agent channel:** audit claim rejections (documentation, eligibility, mis-selling) and strengthen pre-sale checks.

- **Broker channel:** review the policy mix and underwriting standards before growing this channel.
- **Settlement time:** set a 45-day target and track the delayed share monthly.

8. Limitations

- Synthetic practice data.
- 40 claims have no claim amount and 50 policies have no premium. They were not estimated, so total premium is slightly understated. The claim ratio uses approved amount, which is complete.
- 52 claims fall outside their policy period. They were flagged, not removed.
- 85 claims are still pending, so final ratios may change.
- Customer age was invalid or missing for some customers (shown as "Unknown"). Age and channel patterns show association, not cause, and may partly reflect policy mix.
- Claim ratio is only meaningful by policy type, channel, state or age group. It is not shown by claim year, because premium belongs to policies and claims are dated separately.

9. Repository structure

```
insurance-claims-powerbi/
├── data/              raw CSV files
├── sql/
│   ├── 01_cleaning.sql
│   ├── 02_validation.sql
│   └── 03_analysis.sql
├── screenshots/
├── Insurance_Claims_Analysis.pbix
└── README.md
```

10. About

Built by Adarsh as a portfolio project. Skills shown: SQL data cleaning and validation, data modelling, DAX, dashboard design and business storytelling.